American Computer Science League

2022-2023 ● Contest 4: Short Problems Solutions ● Junior Division

1. Graph Theory

The adjacency matrix raised to the 3rd power for paths of length 3 is as follows:

$$\begin{vmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{vmatrix}^3 = \begin{vmatrix} 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 2 & 1 \end{vmatrix}$$

Adding the entries in the right matrix gives 26 paths of length 3.

D. 26

2. Graph Theory

There are 6 cycles from A: ABA, ADA, ADBA, AEDA, AEDBA, ABCDA

E. 6

3. Digital Electronics

The Boolean expression for the digital circuit is: $\overline{A + AB + B}$

$$\begin{aligned} \overline{A + AB + B} &= \overline{A}\,\overline{AB}\,\overline{B} \\ &= \overline{A}\,(\overline{A} + \overline{B})\,\overline{B} \\ &= \overline{A}\,\overline{B} + \overline{A}\,\overline{B} \\ &= \overline{A}\,\overline{B} \end{aligned}$$

D. $\overline{A}\,\overline{B}$

4. Digital Electronics

The following truth table represents the digital circuit:

			X	Y	Z	
A	B	C	Rect(A, B, C)	AX	XC	Y+Z
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	1	1	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0

D. 3

5. What Does This Program Do? (Strings)

Working backwards 6+7+0=13−8=5, 5+6+1=12−8=4, 4+1+1=6, 7+5+0=12−8=4, 3+4+1=8−8=0, 6+3+1=10−8=2, so c=1 and the final string is “1204645”. In other words, this adds two octal numbers character by character.

B. 1204645